

Ivy Hawks

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Education

University of California, San Diego

San Diego, CA

B.S. Data Science

2022-2026

Professional Experience

Student Programmer

Scripps Institute of Oceanography, Lab of David Sandwell, 2022-Present: Manage large data pipelines and processing scripts in Bash, Identify errors and debug issues, manage processes and write workflows to improve efficiency. Train ML models for feature detection, and write tertiary libraries for managing satellite interferometric data in ML pipelines.

Webmaster

UCSD's Biomedical Engineering Society, 2024-2026: Develop from scratch a full stack webserver, managing event timing information, membership within the organization, and dynamic webpages to encourage involvement from the wider student body. Determine optimal database structures, and identify suitable replacement candidates. Train new candidates in using the codebase. Work with leadership to balance financial constraints and use cases.

Projects

Communication Lower Bounds for Multi-Party Graph Traversal

Report: <https://github.com/rybplayer/DSCCapstone/blob/main/q2-report/final-report.pdf>

Many computational problems may be modeled as communication games, in which two or more parties attempt to compute some function with the fewest bits communicated. Here, we present some lower bounds on games pertaining to Graph Ownership, and the traversal problem.

Teletypewriter

Out of a desire for physical media and learning electronics, I am working with a friend to convert electric typewriters into teletype devices. Not only are there electronics problems, but I am also implementing a bespoke driver and communication protocol to learn more about error correction, using my own implementation of the elgamal cryptography protocol.

Skills

- C, Python, Shell scripting, Java
- SQL
- Web Development using Html, CSS, JavaScript, Django
- Machine Learning, with specialty in parametric statistical models
- Unix System Administration
- Graph Theory, Combinatorics, and Linear algebra, with applications to Complexity Theory and Algorithmic Analysis
- Communication of Mathematical concepts to Non-Mathematics technical audience